Atharva Vichare

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SKILLS

Generative & Agentic AI: LLMs (Claude, GPT-4, open-weight), LangChain, LangGraph, RAG Pipelines, Activeloop Deep Lake, Vector Embeddings, Fine-tuning (SFT), Prompt Engineering, Hallucination Control, Ollama

Machine Learning: PyTorch, TensorFlow, PPO / Reinforcement Learning, NLP, Transformer Models, Model Evaluation, Azure Cognitive Services

Languages & Frameworks: Python, TypeScript, JavaScript, C#, Java, React, Node.js, Angular, .NET Core, Three.js, REST APIs

Cloud & DevOps: AWS (Lambda, EC2, Elastic Beanstalk, S3, CloudFront, Route 53, CloudWatch), Azure (AKS, DevOps, Cognitive Services), GCP, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Tools & Data: Claude Code, Cursor, Copilot, MySQL, Vector Databases (Deep Lake), Git/GitHub, JIRA, Microservices, Agile/Scrum

EXPERIENCE

Agentic AI Engineer, Star Communication Inc. – USA 07/2025 – Present

- Designed and shipped a daily-use two-agent LangGraph pipeline (Summarizer + Validator): the Summarizer drafts regulatory briefs and the Validator enforces compliance-rule adherence before human review.
- Implemented hallucination detection by grounding every generated claim against source regulatory text, auto-flagging 22% of drafts for revision and cutting reviewer turnaround time by 35%.
- Built and productionized a RAG service over 5K+ regulatory documents on Activeloop Deep Lake, reaching 85% retrieval relevance via embedding selection, chunking, and retrieval evaluation.
- Containerized the service with Docker and exposed retrieval/generation via TypeScript/Node.js REST APIs; delivered v1 in under two weeks and tuned the pipeline on live reviewer corrections.

AI Research Assistant, University of Delaware – Newark, DE 10/2023 – 07/2025

- Architected conversational AI agents with LLMs, AWS, and React/TypeScript interfaces, generating 400+ hours of training data and improving user engagement 40% through RL optimization.
- Developed 32 Unity virtual environments with multimodal AI (speech-to-text, text-to-speech, visual) and applied PPO reinforcement learning to improve task-completion efficiency 35%.
- Secured Amazon funding for LLM conversational-AI research and contributed to work submitted to ISMAR 2025 on VR-based interactive training systems.

Software Engineer, GEP Worldwide – India 06/2022 – 03/2023

- Built scalable components and AI-enhanced UIs with TypeScript, .NET, AngularJS, Node.js, and Java, boosting application performance 15% for 1K+ daily active users.
- Integrated Azure Cognitive Services and connected deployments to Docker/Kubernetes CI/CD pipelines, cutting deployment time 40%; reduced API response times 30% via a microservices architecture.

Software Engineer, MS Commercial Corporation – India 11/2019 – 11/2021

- Shipped a full-stack React/TypeScript/Node.js/Java application on AWS (EC2, Elastic Beanstalk), increasing site engagement 25%; systematic code reviews reduced post-release defects 30%.
- Owned infrastructure end to end (EC2, S3, CloudFront, Route 53, CloudWatch) with CI/CD, health checks, and rollback; containerized services with Docker/Kubernetes, cutting environment bugs 35%.

OPEN SOURCE

Microsoft Agent Framework github.com/microsoft/agent-framework

- Fixed a reliability defect in the .NET (C#) tool-approval flow by bounding an unbounded auto-approval loop; merged into the main repository after maintainer review and CI (PR #7474, issue #7472).
- Resolved a Python bug causing duplicate function calls on the approval round-trip; merged after maintainer review (PR #7271, issue #7267).
- Two additional Python fixes currently open and under review (PRs #7655, #7635).

PROJECTS

3D Talking Avatar Application github.com/atty57/talking_avatar

- Production-ready 3D conversational AI integrating LLMs with Azure Cognitive Services, React, TypeScript, Three.js, and React Three Fiber; Node.js + Ollama backend enables real-time multimodal interaction with lip-sync accuracy.

CPU Scheduling with Deep Reinforcement Learning github.com/atty57/PPO-for-CPU-Scheduling

- Designed a PPO-based CPU scheduling algorithm achieving a 35% reduction in average waiting time and 20% throughput increase versus traditional scheduling algorithms.

EDUCATION

University of Delaware – Newark, DE — M.S., Computer and Information Sciences 2025

Terna Engineering College – Mumbai, India — B.E., Computer Engineering 2019